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CHAPTER I

INTRODUCTION

1.1 Background of the Study

The advent of advanced automation technologies has the potential. The need for more efficient tools to manage inventory and advancement of automation technologies pushes the emergence of Automated Resource Management Systems in various forms over several decades, evolving from 1960s manufacturing inventory control into the sophisticated, AI-driven platforms used today. Inventory management is critical to running any business, especially in auto supply businesses,where maintaining an optimal stock level directly impacts profitability and customer satisfaction. Inventory management involves tracking, ordering, storing, and selling a company's stock.In the context of auto supply businesses, effective inventory management ensures that essential parts and products are available when needed while avoiding the high costs associated with excess inventory or stockouts. Moreover, the recommendation to adopt a "just-in-time" (JIT) inventory system is insightful, for it can help reduce costs by minimizing the amount of inventory held at any given time, lowering storage costs, and reducing the risk of overstocking. This approach ensures that businesses only order inventory as needed, improving cash flow and preventing excess inventory from becoming obsolete or taking up valuable space.

Because of this people are using web-based applications a lot. These applications are used by institutions and organizations. They make it easy for people to talk to each other, store information and manage data. For example schools using these applications. In many organizations, they still use traditional ways when it comes to managing equipment and resources. They do not use web-based applications for this. Managing equipment and resources is still done by hand. Due to the manual process, problems often arise such as improper recording of information, data loss, slow processing of transactions, and manual data entry leads to version control issues, while disconnected systems create information silos that hinder strategy decision-making.This is the case for organizations that use mobile and web-based applications.Such problems can affect the organization and effectiveness of an institution in managing their resources. Digital technologies and automation systems and data analysis capabilities have brought about significant changes to the global manufacturing sector.The existing. manufacturing systems that depend on human control and centralized management systems cannot achieve current
requirements for flexible production and effective operational processes and delivery of high-quality products and quick market entry.Autonomous manufacturing has become the

primary operational model of the Industry 4.0 framework because it enables production systems to execute self-monitoring and self-analysis and self-optimization with minimal human involvement.The Manufacturing Execution System (MES) has evolved into its present form which functions as the fundamental technology that permits autonomous manufacturing systems to operate without human control.Conventional MES solutions primarily focus on production tracking, scheduling, and compliance reporting.Intelligent MES platforms use artificial intelligence machine learning and real-time data processing to support shop floor decision-making which adapts to changing conditions on the shop floor.Intelligent MES systems create dynamic scheduling systems through operational data analysis which enables them to adapt production schedules and detect anomalies and forecast equipment failures and enhance quality control processes.The intelligent MES system gains better performance through its implementation of cloud-native analytics.Cloud-native systems provide their users with two benefits through elastic computing capacity and permanent system up-time and advanced analytical features which traditional on-site systems cannot deliver. Manufacturers use cloud platforms to process real-time data that comes from different types of machines and enterprise systems.Cloud-native analytics support advanced use cases which include predictive maintenance and cross-plant performance benchmarking and end-to-end supply chain visibility needed for autonomous operations.Smart manufacturing networks use intelligent MES solutions which connect production systems through their continuous operational data exchange with business processes.The system enables manufacturers to achieve real-time operational monitoring through automated decision-making processes which enhance their ability to optimize manufacturing operations and handle unexpected events from inside their facilities and outside their business environment.Organizations need to solve multiple problems which include data protection measures and system operational dependability and employee role preparation and ability to connect different systems to complete the project successfully.The research investigates how intelligent manufacturing execution systems and cloud-native analytics technologies enable manufacturers to achieve self-operating production processes.The research studies technological foundations and architectural designs together with their operational advantages while it identifies major difficulties and prospective future paths.The research investigates how modern manufacturing enterprises can develop self-optimizing production systems which utilize data to establish resilient operational capabilities.The integration of predictive analytics into energy supply chain management is increasingly recognized as a crucial tool for mitigating risks and ensuring operational efficiency. Energy supply chains face numerous challenges, including supply disruptions, fluctuating demand, price volatility, and environmental concerns, which can impact both short-term operations and long-term sustainability. Predictive analytics, leveraging data-driven insights, machine learning algorithms, and statistical models, offers a proactive approach to addressing these challenges by forecasting potential risks and enabling timely interventions.This framework focuses on the application of predictive analytics to identify, assess, and mitigate risks in energy supply chains. Key components of the framework include data collection, analysis of historical trends, real-time monitoring, and forecasting of potential disruptions. By analyzing large datasets from various sources such as market trends, weather patterns, geopolitical factors, and production data, predictive analytics can forecast risks related to energy production, transportation, and consumption, thereby providing valuable insights into potential bottlenecks,

price fluctuations, and demand-supply imbalances.One of the primary advantages of predictive analytics in energy supply chains is its ability to improve decision-making and resource allocation.It enhances risk management by allowing organizations to anticipate and prepare for disruptions before they occur, reducing operational downtime and ensuring a more resilient supply chain.Additionally, predictive models help optimize inventory management, demand forecasting, and supplier relationships, contributing to cost reduction and improved overall efficiency.Despite its potential, the adoption of predictive analytics in energy supply chains faces challenges, such as data quality, technological infrastructure, and the need for skilled professionals to interpret and act on predictive insights.This paper explores these barriers and outlines strategies to overcome them, ensuring the effective implementation of predictive analytics.Ultimately, the framework presented aims to foster a more agile, resilient, and efficient energy supply chain, capable of adapting to emerging risks and contributing to the long-term sustainability of the energy sector.

The dynamic landscape of global supply chains necessitates innovative solutions to tackle challenges in demand forecasting and inventory optimization. Traditional methods, often constrained by limited adaptability and scalability, struggle to manage the complexities of modern supply chains. Artificial Intelligence (AI) has emerged as a transformative force, enabling predictive supply chain management through advanced data analytics, machine learning algorithms, and real-time decision-making capabilities. By harnessing AI-driven tools, businesses can accurately forecast demand patterns, reduce stockouts, and minimize excess inventory, thereby improving operational efficiency and customer satisfaction. AI-powered systems leverage historical data, market trends, and external factors such as economic shifts and weather conditions to provide precise predictions. These tools enhance responsiveness by identifying potential disruptions and enabling proactive measures, ensuring supply chain resilience. Furthermore, AI facilitates seamless integration across supply chain nodes, fostering collaboration and enabling data-driven insights that were previously unattainable.From predictive analytics for demand forecasting to intelligent automation in inventory management, AI-driven tools are revolutionizing the traditional supply chain model. Case studies reveal substantial reductions in holding costs, improved lead times, and enhanced supply chain visibility. However, challenges such as data quality, system integration, and ethical considerations in AI deployment remain critical areas for exploration.The insights presented underscore the pivotal role of AI in driving efficiency and innovation in an increasingly complex and competitive global economy. All markets today are dynamic.Change is in the air everywhere, and change affects strategy. A winning strategy today may not prevail tomorrow. It might not even be relevant tomorrow.There was a time, not too many decades ago, when the world held still long enough for strategies to be put into place and refined with patience and discipline. The annual strategic plan guided the firm.That simply is no longer the case.New products, product modifications,subcategories, technologies, applications, market niches, segments, media, channels, and on and on are emerging faster than ever in nearly all industries—from snacks to fast food to automobiles to financial services to software.Multiple forces feed these changes, including digital technologies, the rise of China and India, trends in healthy living, energy crises, political instability, and more.The result is markets that are not only dynamic but risky, complex, and cluttered.Such convoluted markets make strategy creation and

implementation far more challenging.Strategy has to win not only in today's marketplace but also in tomorrow's, when the customer, the competitor set, and the market context may all be different. In environments shaped by this new. In reality, some firms are driving change. Others are adapting to it. Still others are fading in the face of change.How do you develop successful strategies in dynamic markets? How do you stay ahead of competition? How do you stay relevant to the customer? Implementing an Enterprise Resource Planning system (ERP) in organizations is a strategic move to enhance efficiency and productivity.It discusses how ERP systems streamline processes, centralize data, and facilitate informed decision-making.ERPs eliminate redundancy and improve communication by integrating various functions like finance, HR, and supply chain management. The resulting optimization of resource allocation and reduced manual effort leads to increased productivity.Furthermore, ERP systems enable real-time tracking of key performance indicators, fostering proactive responses to market changes. In summary, ERP deployment empowers organizations to achieve heightened operational efficiency and productivity.

To address these challenges, an Automated Resource Management System can be used that will provide a more systematic way of monitoring, recording, and managing equipment and resources. It offers a lifeline, providing surer footing, a more navigable path, streamline inventory tracking, dramatically improve accuracy, replaces manual spreadsheets with real-time visibility across multiple projects, unified workflows, and provide the firepower that allows companies to respond swiftly to market changes based on deeper insights into consumer behavior and trends. The aim of this study is to develop an automated system that will help facilitate and improve resource management.

1.2 Statement of the Problem

Despite the availability of various technologies that can be used in information management, many organizations will face problems related to manual resource management systems.

This study is going to find out if the Automated Resource Management System really works in making resource management and how the organization runs better. The Automated Resource Management System is what we are looking at. We want to know some things about the Automated Resource Management System.

1. Does the Automated Resource Management System make a difference in how resources are given out and used?

2. Does the Automated Resource Management System make inventory monitoring, control and management better?

3. Does the Automated Resource Management System make tracking and reporting of resources accurate?

4. Does the Automated Resource Management System help reduce delays, mistakes and waste of resources in what the organization does?

5. How better does the Automated Resource Management System make the allocation and use of resources that are available?

6.How well does the Automated Resource Management System manage inventory and the resources of the organization?

7. What kind of effect does the Automated Resource Management System have on how accurate and reliable tracking and reporting of resources are?

8. How much does the Automated Resource Management System help reduce mistakes, delays and waste of resources, in operations?

9. What kind of impact does the Automated Resource Management System have on how productive the organization's how well it performs overall?

Challenges with managing inventory inaccurate inventory tracking, having an excess of unsold merchandise, and insufficient stock to fulfill client needs are the three main issues with inventory management.

- Inadequate or outdated inventory management techniques
- Demand fluctuations caused by customers' changing desires and needs
- Difficulty locating certain commodities in a warehouse
- Depending on the company's field and employer, there are four main kinds of inventory that they may oversee: raw materials, work in progress, finished goods, maintenance, repair, and operations (MRO/Repair) goods

1.3 Objectives of the Study

This study aims to design and develop an Automated Resource Management System that can help in more efficient, orderly, and systematic management of equipment and resources within an organization and to develop an effective system that will streamline the client's business operations and increase efficiency.

To achieve the general objective, this study aims to:

1.To design a system that integrates sales tracking and inventory management with proper categorization and maintaining the data quality during the transaction.
2.To develop a user-friendly web-based system that incorporates a Point of Sale (POS) functionality for efficient inventory management, facilitating faster transactions.
3.To incorporate predictive analytics to compensate for the customer's demand for stocks and prevent overstocking of particular product supplies.
4.To test and evaluate the functionality and accessibility of the proposed system.

1.4 Scope and Limitations

The objective of this project was to create an Automated Resource Management System that could be used to monitor and control resources/equipment at the institution. The system will allow users to login/register, track/monitor resources, store data in a database, and access an administrative dashboard for system administration. The project will only produce a prototype of the Automated Resource Management System for academic purposes; it will not be used in actual production or integrated with other external systems at any other organization.The data collected for the analysis may or may not be the right time for the analysis. The analyzed data may or may provide the accurate results for decision making and the scope of this study focuses on the development and implementation of an automated system, a smart inventory management system designed for mini grocery stores. The system covers core inventory functions such as product registration, stock level monitoring, low-stock alerts, expiration tracking, and the generation of basic inventory reports. It is intended for daily use by store owners, cashiers, and staff with basic computer literacy.

1.5 Definition of Terms

Application Development - We create programs that help companies work better by planning, organizing, testing and keeping an eye on Application Development. The main job of Application Development is to make sure companies can work well by giving employees the tools they need to do their jobs in Application Development. An Automated System is a machine that can work on its own. It usually needs people to supervise the Automated System. The Automated System needs instructions and special tools to work efficiently and accurately in the Automated System. This is important for the Automated System to work properly.

Resource Management - means keeping track of everything a company has, checking if it is working well and making sure it is safe and secure in Resource Management. The people in charge of Resource Management need to know what is going on. This helps prevent things from getting lost or wasted in Resource Management.

Resource Management - is about taking care of the things a company needs to work like buildings and equipment in Resource Management. This includes making sure everything is ready and in condition and getting new things when needed for Resource Management. The people in charge of Resource Management have to control and keep track of everything a company owns from getting supplies to selling things in Resource Management. This helps things run smoothly and makes sure everything is in the place at the right time for Resource Management.

Database - is a collection of files on a computer that stores information about a company. It is like a library where people can find and update information in the Database. The Database can be used for Application Development or Resource Management.

Monitoring System - is a tool that helps companies keep track of how they're managing their resources. It gives them real-time information, which helps them make decisions about Resource Management. We also keep a list of all the equipment a company has including what it is, how much of it we have, what condition it is in and if it is available for Resource Management. This helps us keep track of things and make decisions about what to buy and what to get rid of in Resource Management.

Report Generation - is about making reports that show how well a company is managing its resources in Report Generation. This includes information about equipment, inventory and how things are being used in Report Generation. These reports help leaders make decisions about what to do using Report Generation.

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[2] S., S., & Jogish, D. D. (2023). A Study on Inventory Management.
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[3] Shaikh, A. L. (2024). Review Paper: A Study on Effective Inventory Management and Control.
Source/DOI:
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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The use of automated systems in information management has been associated with improvements in the efficiency and accuracy of organizational processes. Digital systems can help speed up data processing and reduce errors associated with manual operations. The use of intelligent systems and recommendation technologies can also support user experience and trust in information systems.Inventory management is an important part of business operations because it connects purchasing, supply, production, and customer demand. Effective inventory management can help organizations maintain appropriate stock levels while avoiding unnecessary costs associated with excessive inventory. Ineffective inventory management may

also affect the overall performance of an organization [1].Inventory management techniques such as Just-In-Time (JIT) can help organizations reduce inventory holding costs, minimize waste, improve production efficiency, and strengthen coordination with suppliers. Maintaining the necessary level of stock at the appropriate time allows businesses to improve their operational processes.
Studies have also discussed the effects of holding too much or too little inventory. Excess inventory may tie up capital and result in deterioration or obsolescence, while insufficient inventory may lead to stockouts, interruptions in sales, poor customer service, and underutilization of resources [2], [3].

2.2 Local Studies

Local studies related to inventory and resource management emphasize the importance of maintaining accurate records and using appropriate technologies to improve business operations.Inventory management provides organizations with a way to plan resources and properly deploy them during business operations. The integration of different systems can allow separate operations to share information and work together toward common organizational goals. The use of technology in inventory management can therefore support operational efficiency and resource allocation.
The development of automated and technology-supported inventory systems has also been associated with improved monitoring, data management, and decision-making. Modern inventory systems may use automation, data analysis, artificial intelligence, and other technologies to replace or improve manual processes.

2.3 Related Literature

Inventory management is an important component of supply chain management because it affects operational efficiency, customer satisfaction, and profitability. Modern inventory-management approaches include Just-In-Time (JIT), Economic Order Quantity (EOQ), and ABC analysis. These approaches are used to balance inventory costs, stock availability, and resource allocation.Modern inventory systems increasingly use automation, real-time inventory tracking, and predictive analytics to improve inventory control. Inventory management involves monitoring stock, ordering, storing, and managing materials and products. Proper inventory management can help organizations remain organized, respond to changes in demand, and avoid supply-chain problems. Software engineering is also important in developing reliable information systems. According to the literature in the original file, an appropriate software engineering process helps support system quality and reliability. Automation can also contribute to data accuracy, system security, and operational efficiency.
Collaborative Planning, Forecasting and Replenishment (CPFR) is another approach discussed in the literature. CPFR supports cooperation among supply-chain partners and can help improve forecasting accuracy, reduce inventory, and improve supply-chain efficiency.Artificial intelligence and data analysis have also become important technologies in modern inventory systems. These technologies can support automation, resource optimization, inventory management, and business decision-making.

Inventory tracking systems provide organizations with information about the location, movement, and quantity of inventory. These systems may be integrated into inventory-management software and can help organizations monitor available products and stock levels.

2.4 Comparison of Related Approaches

Approach	Main Purpose	Benefit
Traditional/Manual Inventory Management	Recording and monitoring resources manually	Simple to implement
Just-in-Time	Maintaining necessary inventory levels	Reduces holdings costs and waste
CPFR	Collaborative inventory planning and forecasting	Can be forecasting
Automated Inventory Management	Automating inventory and resource processes	Improves monitoring, accuracy, and efficiency

This comparison reflects the approaches discussed throughout the literature in chapter 2 particularly inventory control, CPFR, automation, and technology-supported inventory management.

2.5 Synthesis of the Reviewed Literature

The reviewed studies and literature show that effective inventory and resource management is important for organizational efficiency. Manual processes can make it more difficult to maintain accurate records, monitor resources, and make timely decisions.The literature also shows that automation, information systems, inventory tracking, data analysis, and artificial intelligence can provide organizations with improved ways of managing resources. Automated systems can reduce dependence on manual processes while providing centralized information for monitoring and decision-making.The studies on inventory management, CPFR, automation, artificial intelligence, and information systems support the development of an Automated Resource Management System. The proposed system applies these concepts by providing functions for resource management, inventory monitoring, user management, reporting, and activity tracking.The reviewed byliterature therefore provides a foundation for developing a system that can help organizations manage resources in a more organized and technology-supported manner.

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study used a descriptive and developmental research design to create an Automated Resource Management System. The descriptive part focused on understanding how the organization manages its resources, identifying existing problems, and determining the needs of the users. The developmental part focused on designing, building, testing, and implementing the proposed Automated Resource Management System based on the findings gathered during the

descriptive phase. Data were gathered through interviews, observations, and surveys to obtain information from users and people involved in resource management. The study was guided by the Software Development Life Cycle (SDLC) and Information Systems Theory. These provided a framework for developing the system and ensuring that it addressed the objectives and requirements of the organization.

3.2 System Development Methodology

The study used the Agile Development Model.This model supports collaboration between the developers and stakeholders and allows continuous user feedback throughout the development process.The iterative approach made it possible to identify problems, make adjustments, and improve the system based on user requirements.

1.Requirements Analysis. The researchers identified the needs of the organization by conducting interviews, observing existing processes, and reviewing how resources were managed. The identified problems and requirements served as the basis for the proposed system.

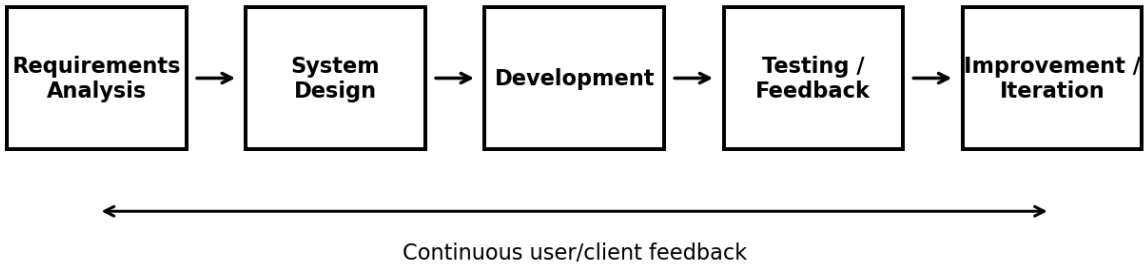
2. **System Design.** The researchers prepared the system structure, database, user interface, process flow, and system architecture. The design aimed to make the system organized, user-friendly, reliable, and suitable for resource management.

3. **Development.** The system was developed in iterative stages. The different modules were built and tested as development progressed. Feedback from users and stakeholders was considered to improve the system.

4. **Testing.** The developed modules were tested to identify errors and verify that the system functions worked correctly. Testing was performed before and after integration of the modules.

5. **Deployment and Maintenance**. After testing, the system was prepared for deployment. Maintenance and user feedback were considered so that errors could be corrected and improvements could be made when necessary.

Figure 3.2 Agile Development Process Used in the Study



3.3 System Architecture

The Automated Resource Management System is a web-based application designed to help organizations manage resources more efficiently. Administrators and authorized users can access the system through a web browser to manage, assign, monitor, and track resources. The system automates resource-related processes, reduces manual errors, and provides information that can support decision-making.The system includes resource assignment and monitoring, inventory management, user and role management, and a dashboard for summaries and reports. A database stores resource and user information and keeps records available to authorized users. Because the system is web-based, users can access it through an internet-connected browser without installing a separate desktop application.

Figure 3.1 Data Flow Diagram (Level 0)



Figure 3.1 illustrates the movement of data between the administrator or authorized user, the Automated Resource Management System, and the database. The user provides inputs or updates, the system processes the information, and the database stores or retrieves the records. The processed information is then returned to the user as results or updated information.

3.4 Tools and Technologies Used

The Automated Resource Management System was developed using web technologies and development tools selected to support an efficient, reliable, and user-friendly application.

Category	Tools / Technologies
Frontend	HTML, CSS, Bootstrap 5
Backend	PHP
Database	MySQL
Development Tool	Visual Studio Code
Local Development Environment	XAMPP

HTML was used to structure the web pages, while CSS was used to improve their appearance. Bootstrap 5 provided ready-made interface components and responsive layouts. PHP was used

for server-side processing and handling user requests. MySQL was used to store and retrieve system data, including resource records and user information. Visual Studio Code was used as the development environment, while XAMPP supported the local development environment.

3.5 Data Gathering Techniques

The researchers used observation and interview as the main data gathering techniques. Through observation, the researchers examined how resources were recorded, monitored, allocated, and maintained. This helped identify existing practices, delays, possible errors, and areas that could be improved through automation.Interviews were conducted with people involved in resource management, such as administrators, staff members, and other relevant users. The interviews helped the researchers understand the users' experiences, needs, problems, and suggestions. The information gathered from observation and interviews served as the basis for identifying system requirements and designing the Automated Resource Management System.

3.6 Testing Procedure

Testing was conducted during system development to identify errors, verify system functions, and determine whether the system met the requirements of its intended users.

Unit Testing. Each individual module or function was tested separately. Examples included resource registration, inventory tracking, user management, and report generation.

System Testing. The integrated system was tested as a whole to determine whether its modules worked together correctly and whether the system performed its intended functions.

User Acceptance Testing (UAT). Administrators and staff members who would use the system tested its functions, usability, and suitability. Their feedback was used to identify improvements and determine whether the system met their needs.

3.7 Evaluation Criteria

The Automated Resource Management System was evaluated based on functionality, usability, reliability, security, and efficiency. These criteria were used to determine whether the system performed its required functions, was easy to use, worked consistently, protected information, and processed tasks efficiently.

Criteria

Functionality

The system should correctly manage resources, track inventory, manage users, and generate reports.

Usability

Users should be able to navigate and use the system without unnecessary difficulty.

Reliability

The system should operate consistently without errors, crashes, or loss of information.

Security

The system should protect information through appropriate authentication, access control, and data protection measures.

Efficiency

The system should process information and complete resource management tasks within an acceptable response time.

CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 System Development and Implementation

The Automated Resource Management System was developed to address the problems identified in the organization's existing manual resource management process. The development started by gathering information from the client and users through communication, observation, and document review. These activities helped identify problems such as difficulty in tracking resources, delayed reporting, and inaccurate inventory records.

The system was then planned based on the identified requirements. The researchers prepared the system structure, database, user interface, process flow, Entity Relationship Diagram (ERD), and Unified Modeling Language (UML) diagrams. These designs served as guides during system development.The system was developed using different modules, including product

management, inventory management, Point of Sale (POS), user management, reports, and activity logs. Each module was developed and tested before being integrated into the complete system.

Figure 4.1 System Development and Implementation Process

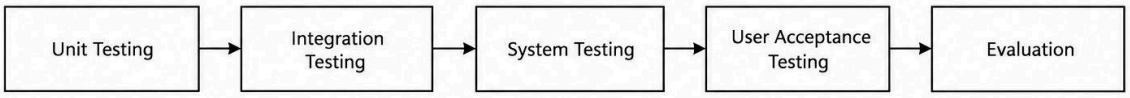


The system was tested through unit testing, integration testing, system testing, and User Acceptance Testing (UAT). Feedback from users was used to identify areas that needed improvement before deployment.

4.2 System Implementation

After successful testing, the system was prepared for implementation in the target environment. The implementation included configuring the database server, installing the application, creating user accounts, and transferring existing resource records when necessary. Users were also provided with orientation and basic training to help them understand the system's features.

Figure 4.2 System Testing and Evaluation



4.3 Testing and Evaluation

Testing was conducted to determine whether the system worked correctly and met the requirements of the organization. Unit testing was used to check individual modules, while integration testing checked whether the modules worked together correctly. System testing examined the overall performance, security, and usability of the application. User Acceptance Testing was also conducted with selected users to determine whether the system met their expectations and operational requirements.The system was evaluated according to Functionality, Usability, Reliability, Security, and Efficiency, which were also identified as the evaluation criteria in Chapter 3.

4.4 Results and Discussion

The testing and user feedback showed that the Automated Resource Management System helped organize resource records and provided a centralized location for storing and retrieving information. Users could monitor resource availability and status without depending entirely on manual records.The system also helped reduce problems associated with manual data entry. The automated updating of information helped minimize duplicate or missing records and made resource information easier to manage. The reporting features provided summaries that could support monitoring and decision-making.The client and users also found the system useful for managing resources. The system provided functions for managing resources, monitoring inventory, managing users, generating reports, and recording activities.These results indicate that the developed system addressed the resource management problems identified during the earlier stages of the study.

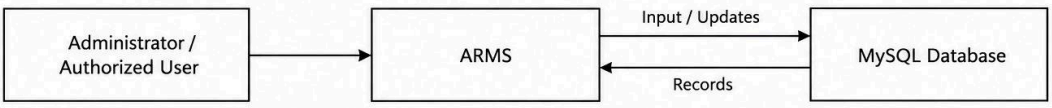
4.5 Hardware and Software Requirements

The system requires appropriate hardware and software to operate properly. The original study specifies a minimum Intel Core i3 processor, 4 GB RAM, 256 GB SSD, 1366 × 768 display, and 10 Mbps internet connection, with higher specifications recommended for better performance. It also considers Android 10 as a supported mobile environment, with Android 13 or higher recommended.The software environment includes HTML, CSS, JavaScript, PHP, MySQL, XAMPP, Visual Studio Code, and modern web browsers. These technologies support the system's interface, server-side processing, database management, local development, and browser-based access.

4.6 Database Design

The database serves as the central storage component of the system. It stores information related to products, inventory, transactions, user accounts, and other records required by the application. MySQL was used as the relational database management system.The database is connected to the PHP-based application, allowing authorized users to add, retrieve, update, and manage information. The database structure supports user management, inventory management, product management, transactions, and reporting.

Figure 4.3 System Database Structure



CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

The Automated Resource Management System was developed and evaluated to address the problems encountered in manually managing organizational resources. The findings were based on the system testing, observations, and feedback from the client and users.

The major findings of the study are:

- Organized Resource Records. The system provided an organized way of recording and storing resource information, making records easier to access and manage.
- Improved Resource Monitoring. The system allowed authorized users to view the status and availability of resources without manually checking numerous files or records.
- Reduced Recording Errors. The automated recording and updating of information helped reduce duplicate, missing, and incorrect records caused by manual data entry.
- Efficient Resource Allocation. The system helped the organization monitor available and currently used resources, supporting better decisions when allocating resources.
- Reports and Information. The reporting features provided summaries of resource information that helped the client monitor resources and support decision-making.
- Security Controls. The system included controls that limited access to authorized users and protected resource-related information.
- Ease of Use. Based on the users' feedback and observations, the system was understandable and allowed users to perform common tasks more easily.
- Addressed Client Problems. The system addressed the problems identified in the manual process by providing a more organized, faster, and reliable way of managing resources. Chapter 1-5 updated .pdf

5.2 Conclusions

Based on the findings, the Automated Resource Management System achieved its intended purpose of improving the management, monitoring, and tracking of organizational resources.The system provided a centralized way to record, update, and retrieve resource information. It also helped reduce the problems associated with manual record-keeping, such as difficulty in finding information, possible recording errors, and delayed monitoring.The integration of the dashboard, inventory management, user management, reports, and activity logs provided the client with a more structured way of handling resource-related activities. The results from the

system evaluation showed that the system addressed the identified needs of the organization and met the expectations of the client and users.

5.3 Recommendations

Based on the findings and feedback from the client and users, the following recommendations are proposed:

- **Improve the Dashboard and User Interface.** The dashboard and interface may be further improved based on user feedback to make navigation clearer and easier.
- **Improve Reporting and Exporting.** The reporting feature may be expanded to allow users to export reports in formats such as PDF, Excel, or CSV.
- **Strengthen Security.** User authentication, password protection, and role-based access may be further improved to provide stronger protection for resource information.
- **Add Notifications and Alerts.** Notifications may be added for low stock, overdue resources, and other important resource-related activities.
- **Improve Activity Logging.** The activity log may be enhanced to provide a clearer record of user actions, helping administrators monitor changes and maintain accountability.
- **Maintain a Simple Interface.** Future improvements should continue to prioritize a simple and user-friendly interface, especially for users who have limited computer experience.
- **Integrate with Other Systems.** Future researchers may explore connecting the Automated Resource Management System with other organizational systems, such as accounting or employee management systems, to create a more integrated system.